GOBI M350 Fieldbook

APPN Aerial SOP — Locked revision 1.0

APPN – GOBI M350 Fieldbook

For a high-level description of the GOBI sensor package and how it sits alongside the other APPN platforms, see the Platforms Overview.

For the GOBI fieldbook covering the Inspired Flight IF1200A, see GOBI IF1200 Fieldbook.

This fieldbook provides a standardised operational guide for APPN GOBI UAV deployments **on the DJI M350**, supporting safe flight operations, consistent sensor configuration, and high-integrity data capture. It is intended for trained APPN staff conducting hyperspectral, LiDAR, RGB, and GNSS-INS data acquisitions, and promotes repeatability, transparency, and confidence in downstream data analysis across APPN operations.

> **Important**
>
> **This protocol must be followed for all standard APPN GOBI M350 UAV flights.** Adherence to these procedures is essential to ensure operational safety, data integrity, and comparability of datasets across deployments. For any flights that **fall outside standard operating procedures**, detailed records must be kept documenting all deviations, including the specific settings changed, the rationale for those changes, and any anticipated implications for data quality or analysis.

Document Structure

1. Equipment Checklist — what to bring to the field.
2. Preflight Planning — survey / capture polygons and DJI Pilot 2 flight planning.
3. Onsite Preflight Operations — weather and airspace checks, panel and GCP layout, aircraft setup.
4. GOBI Sensor Configuration — sensor power-up, exposure, dark reference, and mission start.
5. Flight Operations — takeoff, dynamic alignment, autonomous mission, and landing.
6. Post-Flight Sensor Configuration — stopping the mission and confirming data on the sensor.
7. Data Confirmation — expected file counts and sizes by sensor.
8. Post-Flight — pack-down of aircraft, payload, and ground reference kit.
9. Offload Data — downloading from the sensor and storing under the APPN folder structure:

- Data storage, processing & validation

10. Appendix — FTP & service reference, Headwall polygon tool workflow, static IP setup, and IP address reference.

Equipment Checklist

> **Note**
>
> Ensure batteries for all equipment are fully charged before heading to the field. Ensure charging cables are available for necessary equipment.

☐ **Aircraft**
☐ DJI M350
☐ Aircraft batteries and spares
☐ Landing gear
☐ Landing pad
☐ Spare parts
☐ Tools
☐ Logbook
☐ **Radio Control Transmitter / Ground Control Station**
☐ **GRYFN Gobi**
☐ Gobi system (power once each month to charge the RGB camera internal battery)
☐ Power cables, chargers
☐ Ethernet cable (and dongle if needed)
☐ SD card
☐ *Capture* and *survey* polygon files
☐ **Ground reference kit**
☐ 2 x Reflectance calibration panels (11%, 30%, 56%, 82%)
☐ 1 x Calibration validation panels (20%, 45%)
☐ 5 × Propeller Aeropoints or other ground control points (GCPs) and/or RTK GNSS system
☐ 2 × folding tables to elevate panels
☐ *If over 50 km from CORS base station*, a portable RTK base station (link to GRYFN gitbook)
☐ **Accessories**
☐ Safety gear (signage and high-vis vests)
☐ Aeronautical radio
☐ Field laptop and spare batteries
☐ External storage media
☐ Water, food, esky, sunscreen, bug spray, first aid kit, etc.
☐ Spirit bubble, spirit level (or angle measurement) and measuring tape
☐ Portable fan (for cooling GOBI during data offload)
☐ External power brick (for charging UAV RC)

> **Note**
>
> The **Ground reference kit** above assumes the Standard Dual ELM panel flight design. Other flight designs may require additional equipment (e.g. extra reflectance panel sets, additional GCPs and folding tables for multi-flight captures, or extra targets for validation flights). See the Standard Flight Procedure and the Validation Flight Procedure for the full equipment requirements of each design.

Preflight Planning

> **Important**
>
> Ensure that you apply for UAV flight approvals for locations and dates of flights well in advance. Further planning documentation for the GOBI can be found in the GRYFN operations gitbook.

1. Using a GPS survey system (Emlid, Trimble…) or GIS software, create a polygon with a 5 m margin around the area of interest. Make sure the polygon includes the areas where you will place your calibration panels and GCPs, with an additional 5 m buffer to avoid incomplete data.
2. Create two polygons — a *survey* polygon and a *capture* polygon. Ensure the *survey* polygon is 2× the flight speed larger than the *capture* polygon in the direction of travel (as per the figure below). Save both polygons uniquely as KML files using the convention `YYYYMMDD_XXXX_capture` and `YYYYMMDD_XXXX_survey` (`YYYY` = year; `MM` = month; `DD` = day, must be 2 digits; `XXXX` = a short reference or abbreviation for the job).

3. *If using QGIS*, import the *capture* KML into Google Earth and immediately export it again. This is due to the required formatting Google Earth supplies, versus QGIS.
4. Import the *capture* KML into the HPI Polygon Tool and export. This polygon sets the activation bounds of the hyperspectral sensor.
5. Load the *survey* KML onto a USB stick or microSD card.
6. Power on the controller. Open DJI Pilot 2 and select **Flight Route → + Import Route (KMZ/KML)**. Alternatively, you can create an Area Route on the controller. *Note that you do not need the UAV powered on for this, just the controller.*
7. Using both DJI Pilot 2 and the GRYFN flight calculator, determine the speed, altitude, and frame period required to survey the area of interest.
8. Ensure the frame period is **> 20% oversampled (30% is our default) and the side overlap is 40%**. In windier conditions (> 5 m/s), increase the side overlap to 50%.

Onsite Preflight Operations

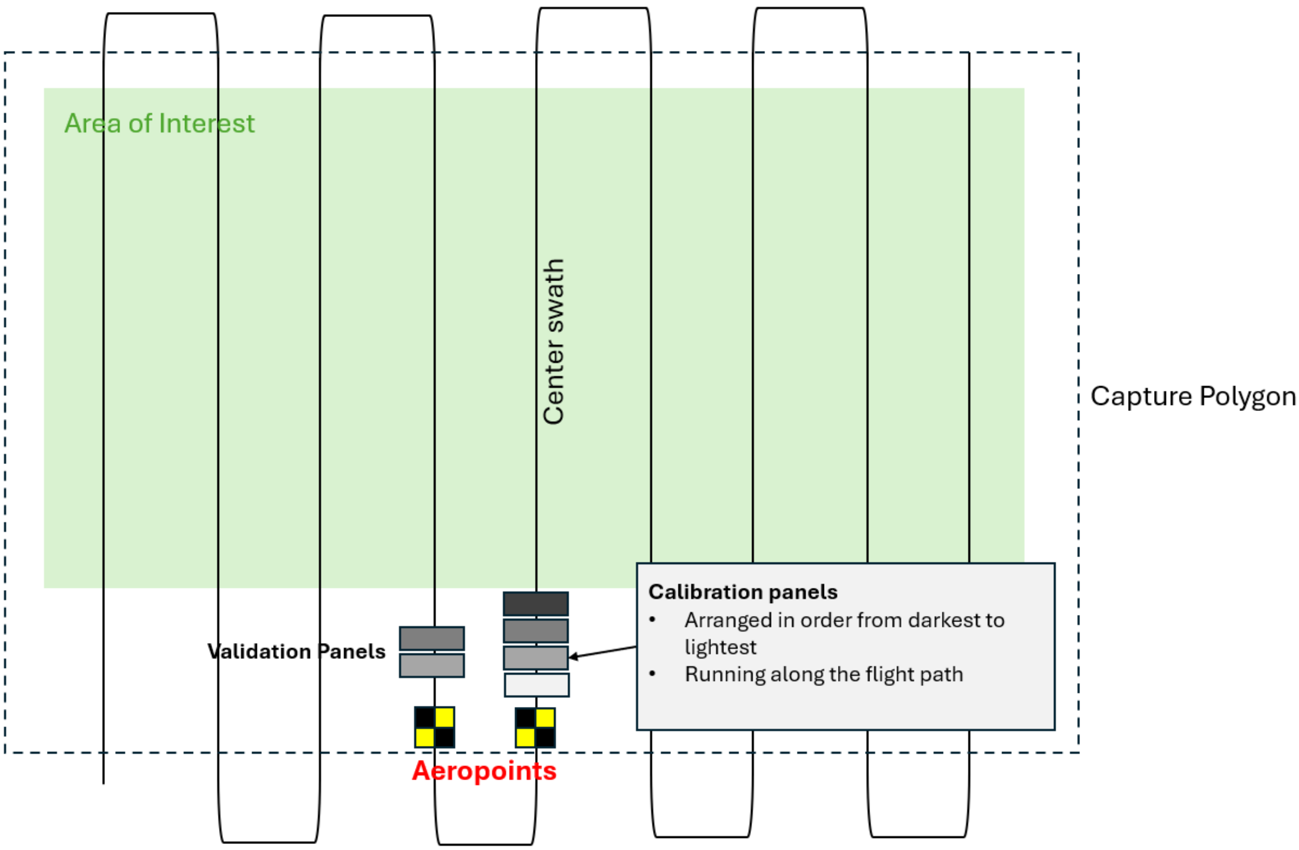
1. Conduct airspace and weather checks.

- No cloud cover.
- Maximum wind speed for quality hyperspectral capture is 6 m/s (22 km/h). Any wind speed over 5 m/s (18 km/h) should be recorded.
- Do not operate in conditions below 0 °C or above 40 °C.
- Try to ensure the sun is within ±20° of solar noon (approximately 2 hours before or after noon). **However**, *this depends on time of year and latitude — check NOAA solar calculator if unsure.*

2. Turn on the aeronautical radio and set to local CTAF (find in ERSA).
3. *Optional, if over 50 km from a CORS station:* set up the Emlid RTK (install a peg or permanent GCP below the base station for recurring flights), let it run for at least 15 minutes, and start recording the RINEX file before flying.
4. Set up reflectance targets for calibration and validation, using a spirit bubble to ensure they are all laid flat. Place 2 GCPs in the field to ensure geometric calibration. GCPs should be located next to calibration panels,

ensuring they are in alternate flight lines.

- Both the calibration and validation panels should be placed on level folding tables to avoid dirt, reduce angle, and reduce spectral spillover. Keeping them level is important.
- 2 sets of GRYFN calibration reflectance panels should be used if available, especially for flight lengths > 15 minutes.
- When using **1** calibration panel, the panel should be placed in the centre of a flight line in the middle of the flight.
- When using **2** panels, they should be placed in the centre of flight lines 1/3 and 2/3 of the mission, noting that all missions would be less than 30 minutes long due to limitations of the IF1200.
- You will need to move the panels if you will be flying multiple missions. Panels need to be in every flight in multi-flight mission capture so that you fly over a calibration panel approximately every 15 minutes.



5. Set up the landing pad and UAV in a safe RTH location.

- In dusty environments an additional tarp should be used under the landing pad.

6. On the controller, set up a safe UAV RTH location, RTH altitude, and other geo-fencing settings.
7. Attach the GOBI payload to the aircraft:

- Check that all lenses and sensors are clean. If not, use professional lens cleaning wipes to clean them.
- Connect the power cable from aircraft to payload (non-standard payload bus only).
- Connect both GNSS antenna cables to the correct ports (match A1 and A2).
- **Remove RGB and Nano HP lens caps.**
- Insert the RGB SD card.
- Connect the C-type power cable to power the gimbal (make sure the B side of the cable is toward the outer side of the drone and the upper side of the light sensor).
- *Optional, in warm ambient conditions:* activate the battery-powered fan. Avoid leaving GOBI powered on in stagnant air or high heat (> ~35–38 °C).

8. Power on the radio controller; check battery status.
9. Launch DJI Pilot 2 on the M350 controller.

10. Review the flight plan, checking operational height and double-checking that area of interest, GCPs, and reflectance panels are all within the capture polygon.
11. Power on the aircraft; confirm connection to RC/GCS and battery status.
12. If you are using RTK, check that you are receiving GNSS corrections. There should be more than 8 satellites for good correction (check the RTK fix in the M350 controller settings).

GOBI Sensor Configuration

1. Place the 80% exposure reference panel under the Nano HP lens; avoid casting shadows.

- Place the drone legs upon exposure-angle cones, ensuring a fixed angle is achieved each exposure setting.
- The maximum distance from the VNIR HS sensor to the GOBI reflectance panel is 40 cm. This is calculated based on the FOV and a 20 cm panel size.

2. Connect to GOBI Wi-Fi or connect the ethernet cable to the payload. *If using ethernet, ensure a static IP address for the sensor at* `10.0.65.2`.
3. Navigate to the GOBI WebUI at `10.0.65.50`.
4. Ensure a valid altitude is shown in the top-right corner of the WebUI (~10–20 m error is acceptable). Typically, this will normalise to ground elevation within 4–6 minutes.
5. Name the mission, using the convention `YYYYMMDD_XXXX` (`YYYY` = year; `MM` = month; `DD` = day, must be 2 digits; `XXXX` = a short reference or abbreviation for the job).
6. Ensure the correct UAV (M350) is selected in the dropdown box. **This is very important.**

> **Caution**
>
> Selecting the incorrect drone will render all data captured unrecoverable

7. Scroll down to polygon settings. Ensure triggering is set by **Polygon** in the dropdown menu. Set min and, optionally, max altitude triggers (recommended to set a minimum of at least 20 m to prevent unnecessary data capture).
8. Import the polygon KML or select from previously flown missions using the dropdown menu. Inspect the map to ensure it aligns with expectations.
9. Return to the top of the WebUI, open the Flight Calculator, and update flying height / flight speed (default 40 m height, default speed 3 m/s); adjust the oversampling buffer (recommended setting is 30%). Frame period will be automatically updated.
10. Adjust exposure until ~95% spectral saturation at the **peak of the 90th percentile line** without exceeding the frame period. Use the lowest gain mode that still provides sufficient saturation.
11. Place the lens cap on the Nano HP.
12. Click **Collect Dark Reference**; wait to finish.
13. **Remove the lens cap.**
14. Remove the exposure reference panel and return it to its case.
15. Press **Start Mission**.
16. Remove the ethernet cable (if using) and begin flight operations.

Flight Operations

1. Ensure the aircraft is in **Position** flight mode.
2. Clear people/objects away from the UAV.
3. Notify crew/observers that takeoff is beginning.
4. Begin manual takeoff, check stick controls work, and then fly to ~12 m AGL; **do not exceed** the trigger altitude (20 m default) before mission start.
5. Perform dynamic alignment (one to two figure-8 patterns) at a lower height than capture height and outside of the capture polygon area.
6. Enable autonomous mission.
7. Monitor UAV battery voltage/percentage in flight.

8. After mission, switch back to **Position** mode to regain manual control.
9. Lower to below capture height and perform post-mission dynamic alignment (figure-8).
10. Land and disarm UAV. Be careful when landing the M350 — downward sensors are blocked when using GOBI, so bring it in slowly.
11. Leave the UAV, transmitter, and sensor powered on; begin post-flight checks. Ensure data transfer is finished before turning off the sensor.

> **Warning**
>
> **There is no hot-swap protection on the GOBI.** Do not mate or de-mate the Smart Dovetail while the aircraft is powered.

> **Warning**
>
> Hot-swapping batteries is **not recommended** with any high-end GNSS solution. Static data can cause IMU drift and degrade post-processed trajectory quality. **Treat each flight as a new flight.**

Post-Flight Sensor Configuration

1. Reconnect to GOBI Wi-Fi or ethernet cable.
2. Open a browser to `10.0.65.50` or refresh the UI.
3. Press **Stop Mission**.
4. Replace the Nano HP and RGB lens caps.
5. In the WebUI, ensure all data has captured correctly (see the Data Confirmation table below).

Data Confirmation

Sensor type	File count	File size	Notes
GNSS-INS	1 file	~1.5 MB per minute	A new file will be created if time rolls ov
LiDAR	1 per minute	~175 MB per file	First & last file may be smaller. Files ma
VNIR	Mission time (s) / Frame Period value ~ number of data cubes	Several GB even for short flights.	Check that `frameindex`, `imu_gps`, and `sett`
RGB	One image every 2 s* (by default)	~30–70 MB per image	Check the event file vs the number of ima

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Post-Flight

1. Power off the drone, then the controller. Inspect the drone for any damage and report for maintenance if necessary.
2. Disconnect the ethernet cable from the payload, if using.
3. Disconnect the aircraft batteries.
4. Disconnect the payload power cable.
5. Disconnect the payload GNSS cables.
6. Undo the mounting clamp, remove the payload, and place it in its case.
7. Pack up the drone and all gear from site, double-checking against the checklist at the start of this document to ensure everything is accounted for.

Offload Data

1. Open WinSCP, FileZilla, or similar FTP software.
2. Connect to the GRYFN Gobi via `gryfn@10.0.65.50` (password: `gryfn`).
3. Raw GNSS & LiDAR PCAP location: `/data/{mission name}`.
4. VNIR data location: `/data/capturedData/captured/{dateTime}`.

5. GOBI logs location: `/data/gryfn.log.{date}`.
6. RGB images are stored on the camera SD card.
7. Download GNSS, LiDAR, logs, and RGB after each mission; download hyperspectral data later due to its size.
8. Clear data directories after download and backup to avoid filling the 500 GB SSD.

Data storage, processing & validation

All paths below follow the APPN folder structure. Formal paths use the wiki's placeholder syntax; an example follows each.

1. Downloaded data should be stored in the correct `T0_raw` folder.

Formal path:

```
./{Node}/
{YYYY_ProjectDesc[_I|E][_Researcher][_org]}/
{YYYYSiteName[_F|C]}/
{SensorPlatform}/{YYYYMMDD}/run_XX/T0_raw/
```

Example:

```
./USYD_Narrabri/2025_SIFCal/2025IAWatson_F/GOBI/20250825/run_00/T0_raw/
```

2. Data from the GCP points should be saved in the `T0_raw/Vault` folder (location may change).

Formal path:

```
./{Node}/
{YYYY_ProjectDesc[_I|E][_Researcher][_org]}/
{YYYYSiteName[_F|C]}/
{SensorPlatform}/{YYYYMMDD}/run_XX/T0_raw/Vault/
```

Example:

```
./USYD_Narrabri/2025_SIFCal/2025IAWatson_F/GOBI/20250825/run_00/T0_raw/Vault/
```

3. Important information and issues should be recorded in the `FieldNotes.txt` file.

Formal path:

```
./{Node}/
{YYYY_ProjectDesc[_I|E][_Researcher][_org]}/
{YYYYSiteName[_F|C]}/
{SensorPlatform}/{YYYYMMDD}/FieldNotes.txt
```

Example:

```
./USYD_Narrabri/2025_SIFCal/2025IAWatson_F/GOBI/20250825/FieldNotes.txt
```

4. Per-run information (e.g. APEx test conditions) should be stored in `RunOverview.csv`.

Formal path:

```
./{Node}/
{YYYY_ProjectDesc[_I|E][_Researcher][_org]}/
{YYYYSiteName[_F|C]}/
{SensorPlatform}/{YYYYMMDD}/RunOverview.csv
```

Example:

```
./USYD_Narrabri/2025_SIFCal/2025IAWatson_F/GOBI/20250825/RunOverview.csv
```

> **Important**
>
> When data from a failed run is being kept (e.g. debugging with GRYFN), the `RunFailed` boolean column of `RunOverview.csv` must be set to `True`.

5. The bundled `.graw` should be saved in the same `T0_raw` folder.
6. The `.gpro` should be generated using the APPN GPT Pipeline (`.json` and wiki links *TODO: add links*) and stored in the adjacent `T1_proc` folder.

Formal path:

```
./{Node}/
{YYYY_ProjectDesc[_I|E][_Researcher][_org]}/
{YYYYSiteName[_F|C]}/
{SensorPlatform}/{YYYYMMDD}/run_XX/T1_proc/
```

Example:

```
./USYD_Narrabri/2025_SIFCal/2025IAWatson_F/GOBI/20250825/run_00/T1_proc/
```

7. Standard QA process should be performed following this guide.

Appendix

FTP & Service Reference

Service	Protocol	IP	Port	User	Password
GOBI	FTP	`10.0.65.50`	22	`gryfn`	`gryfn`
SBG	FTP	`10.0.65.100`	21	`operator`	*(none)*
Headwall polygon	HTTP	`apps.headwallphotonics.com`	—	—	—

Headwall Polygon Tool — Workflow

1. Import KML.

- Must be in Google Earth format.

2. Export KML.
3. Save KML.
4. In File Explorer, rename the KML as the survey name.

Setting a Static IP Address

If connecting over an ethernet connection, a static IP address will need to be set:

1. Open *Network and Internet Settings*.
2. Open *Change Adapter Options*.
3. Open *Properties* for the Ethernet port/adapter.
4. Open *Properties* for IPv4.
5. Set IP address to `10.0.65.2`.
6. Set subnet mask to `255.255.255.0`.
7. Press **OK**.

IP Address Reference

Service	Address
GRYFN WebUI	`10.0.65.50`
HSInsight	`10.0.65.50:8080`
Ouster	`10.0.65.128`
SBG	`10.0.65.100`